

clipper—1-clipper分析

1.clipper环境构建,之前详细过程在我的知乎上写了, 配上网址[!!! 血书clipper安装过程 - 知乎](#)

2.其他软件的安装借助conda

```
##所需安装软件:
cutadapt
umi_tools
STAR
gatk
wf_get_reproducible_eclip_peaks
```

3.clipper软件环境及版本号

```
# packages in environment at /home/gongweikang/.conda/envs/clipper3:

#

# Name                                Version                                Build Channel
_libgcc_mutex                         0.1                                    main conda-forge
alabaster                             0.7.13                                pypi_0 pypi
babel                                 2.14.0                                pypi_0 pypi
bedtools                              2.29.2                                hc088bd4_0 bioconda
blas                                  1.0                                    mk1
bowtie2                               2.4.2                                py37h8270d21_1 bioconda
bz2file                               0.98                                  py37h06a4308_1
bzip2                                 1.0.8                                  h7b6447c_0
ca-certificates                      2024.11.26                            h06a4308_0
certifi                               2022.12.7                            py37h06a4308_0
charset-normalizer                    3.4.1                                pypi_0 pypi
clipper                              2.0.0                                pypi_0 pypi
curl                                  7.71.0                                hbc83047_0
cutadapt                              2.6                                  py37h516909a_0 bioconda
cwl                                    0.0.1                                pypi_0 pypi
cycler                                0.10.0                                py37_0
cython                                0.29.20                              py37he6710b0_0
dbus                                  1.13.16                              hb2f20db_0
deeptools                             3.5.4.post1                          pypi_0 pypi
deeptoolsintervals                   0.1.9                                py37hf01694f_2 bioconda
dnaio                                  0.3                                  py37h14c3975_1 bioconda
docutils                              0.19                                  pypi_0 pypi
expat                                  2.2.9                                he6710b0_2
fontconfig                             2.13.0                                h9420a91_0
fonttools                             4.38.0                                pypi_0 pypi
freetype                               2.10.2                                h5ab3b9f_0
future                                0.18.3                              py37h06a4308_0
gatk                                   3.6                                    5 bioconda
gatk4                                 4.0.5.1                              0 bioconda
glib                                   2.65.0                                h3eb4bd4_0
gst-plugins-base                      1.14.0                                hb2d80ab_1
gstreamer                             1.14.0                                hb31296c_0
htseq                                  0.11.3                              py37hb3f55d8_0 bioconda
```

|                    |              |                |             |
|--------------------|--------------|----------------|-------------|
| icu                | 58.2         | he6710b0_3     |             |
| idna               | 3.10         | pypi_0         | pypi        |
| imagesize          | 1.4.1        | pypi_0         | pypi        |
| importlib-metadata | 6.7.0        | pypi_0         | pypi        |
| intel-openmp       | 2020.1       | 217            |             |
| java-jdk           | 8.0.92       | 1              | bioconda    |
| jinja2             | 3.1.5        | pypi_0         | pypi        |
| joblib             | 0.15.1       | py_0           | conda-forge |
| jpeg               | 9b           | h024ee3a_2     |             |
| kiwisolver         | 1.2.0        | py37hfd86e86_0 |             |
| krb5               | 1.18.2       | h173b8e3_0     |             |
| ld_impl_linux-64   | 2.33.1       | h53a641e_7     | conda-forge |
| libcurl            | 7.71.0       | h20c2e04_0     |             |
| libdeflate         | 1.0          | h14c3975_1     | bioconda    |
| libedit            | 3.1.20191231 | h7b6447c_0     |             |
| libffi             | 3.3          | he6710b0_1     |             |
| libgcc             | 7.2.0        | h69d50b8_2     | conda-forge |
| libgcc-ng          | 9.1.0        | hdf63c60_0     |             |
| libgfortran-ng     | 7.3.0        | hdf63c60_0     |             |
| libpng             | 1.6.37       | hbc83047_0     |             |
| libssh2            | 1.9.0        | h1ba5d50_1     |             |
| libstdcxx-ng       | 9.1.0        | hdf63c60_0     |             |
| libuuid            | 1.0.3        | h1bed415_2     |             |
| libxcb             | 1.13         | h1bed415_1     |             |
| libxml2            | 2.9.10       | he19cac6_1     |             |
| markupsafe         | 2.1.5        | pypi_0         | pypi        |
| matplotlib         | 3.5.3        | pypi_0         | pypi        |
| mk1                | 2020.1       | 217            |             |
| mk1-service        | 2.3.0        | py37he904b0f_0 |             |
| mk1_fft            | 1.1.0        | py37h23d657b_0 |             |
| mk1_random         | 1.1.1        | py37h0573a6f_0 |             |
| ncurses            | 6.2          | he6710b0_1     |             |
| numpy              | 1.18.5       | py37ha1c710e_0 |             |
| numpy-base         | 1.18.5       | py37hde5b4d6_0 |             |
| numpydoc           | 1.5.0        | pypi_0         | pypi        |
| openjdk            | 8.0.152      | h7b6447c_3     |             |
| openssl            | 1.1.1w       | h7f8727e_0     |             |
| packaging          | 24.0         | pypi_0         | pypi        |
| pandas             | 1.0.5        | py37h0573a6f_0 |             |
| pcre               | 8.44         | he6710b0_0     |             |
| perl               | 5.26.2       | h14c3975_0     |             |
| pigz               | 2.6          | h27cfd23_0     |             |
| pillow             | 9.5.0        | pypi_0         | pypi        |
| pip                | 20.1.1       | py37_1         |             |
| plotly             | 5.9.0        | py37h06a4308_0 |             |
| py2bit             | 0.3.0        | py37hf01694f_4 | bioconda    |
| pybedtools         | 0.8.1        | py37h4ef193e_1 | bioconda    |
| pybigwig           | 0.3.17       | py37hfb91e9a_1 | bioconda    |
| pygments           | 2.17.2       | pypi_0         | pypi        |
| pyparsing          | 2.4.7        | py_0           |             |
| pyqt               | 5.9.2        | py37h05f1152_2 |             |
| pysam              | 0.15.3       | py37hda2845c_1 | bioconda    |
| python             | 3.7.7        | hcff3b4d_5     |             |
| python-dateutil    | 2.8.1        | py_0           | conda-forge |
| python_abi         | 3.7          | 1_cp37m        | conda-forge |
| pytz               | 2020.1       | py_0           |             |

|                               |           |                |             |
|-------------------------------|-----------|----------------|-------------|
| qt                            | 5.9.7     | h5867ecd_1     |             |
| readline                      | 8.0       | h7b6447c_0     |             |
| regex                         | 2022.3.15 | py37h7f8727e_0 |             |
| requests                      | 2.31.0    | pypi_0         | pypi        |
| samtools                      | 1.7       | 1              | bioconda    |
| scikit-learn                  | 0.23.1    | py37h423224d_0 |             |
| scipy                         | 1.5.0     | py37h0b6359f_0 |             |
| setuptools                    | 47.3.1    | py37_0         |             |
| sip                           | 4.19.8    | py37hf484d3e_0 |             |
| six                           | 1.15.0    | py_0           |             |
| snowballstemmer               | 2.2.0     | pypi_0         | pypi        |
| sphinx                        | 5.3.0     | pypi_0         | pypi        |
| sphinxcontrib-applehelp       | 1.0.2     | pypi_0         | pypi        |
| sphinxcontrib-devhelp         | 1.0.2     | pypi_0         | pypi        |
| sphinxcontrib-htmlhelp        | 2.0.0     | pypi_0         | pypi        |
| sphinxcontrib-jsmath          | 1.0.1     | pypi_0         | pypi        |
| sphinxcontrib-qthelp          | 1.0.3     | pypi_0         | pypi        |
| sphinxcontrib-serializinghtml | 1.1.5     | pypi_0         | pypi        |
| sqlite                        | 3.32.3    | h62c20be_0     |             |
| star                          | 2.7.10b   | h9ee0642_0     | bioconda    |
| tbb                           | 2020.3    | hfd86e86_0     |             |
| tenacity                      | 8.0.1     | py37h06a4308_1 |             |
| threadpoolctl                 | 2.1.0     | pyh5ca1d4c_0   | conda-forge |
| tk                            | 8.6.10    | hbc83047_0     |             |
| tornado                       | 6.0.4     | py37h7b6447c_1 |             |
| typing-extensions             | 4.7.1     | pypi_0         | pypi        |
| umi_tools                     | 1.0.1     | py37hf01694f_2 | bioconda    |
| urllib3                       | 2.0.7     | pypi_0         | pypi        |
| wheel                         | 0.34.2    | py37_0         | conda-forge |
| xopen                         | 0.7.3     | py_0           | bioconda    |
| xz                            | 5.2.5     | h7b6447c_0     |             |
| zipp                          | 3.15.0    | pypi_0         | pypi        |
| zlib                          | 1.2.11    | h7b6447c_3     |             |

4. 从 FASTQ 文件中提取 Unique Molecular Identifiers (UMIs)。UMIs 是在文库制备过程中添加到每个原始 DNA 或 RNA 分子上的短的随机核苷酸序列，用于区分 PCR 扩增产生的重复序列和原始分子。通过使用 UMIs，可以在后续的数据分析中识别并去除 PCR 引入的偏差，从而提高测序数据的质量和准确性。

```
umi_tools extract --stdin ${n}_1.fq.gz --bc-pattern NNNNNNNNNN --log
${n}_processed.log --stdout ${n}_umi.fq
```

5. 去除特定的接头 (adapter) 和引物序列，并根据设定的质量和长度标准来过滤读段

```
cutadapt --match-read-wildcards --quality-cutoff 6 --times 1 -e 0.1 -o 1 -m
18 -a NNNNNAGATCGGAAGAGCACACGTCTGAACTCCAGTCAC -g CTTCCGATCTACAAGTT -g
CTTCCGATCTTGGTCCT -o ${n}_umi_trim.fq ${n}_umi.fq > ${n}.metrics
```

6. 比对重复序列原件，取出未必对上的读段

```
STAR --runMode alignReads --runThreadN 8 --genomeDir ${index} \
    --genomeLoad LoadAndRemove --readFilesIn ${n}_umi_trim.fq \
    --outSAMunmapped within --outFilterMultimapNmax 30 --
outFilterMultimapScoreRange 1 \
    --outFileNamePrefix ${n} \
    --outSAMattributes All --outStd BAM_Unsorted --outSAMtype BAM Unsorted \
    --outFilterType BySJout --outReadsUnmapped Fastx --outFilterScoreMin 10 -
-outSAMattrRGline ID:foo --alignEndsType EndToEnd
```

7. 未必对上的读段比对全基因组进行重新比对（因为我们还要特定病毒序列，因此这一步要分两步，同步进行

#比对全基因组序列（由于后续clipper软件原因需要下载ucsc上的hg19序列，其他的不满足要求，会出错无法进行）

```
STAR --runMode alignReads --runThreadN 9 --genomeDir ${index} \
    --genomeLoad LoadAndRemove --readFilesIn ${n}.out.mate1 \
    --outSAMunmapped within --outFilterMultimapNmax 1 --
outFilterMultimapScoreRange 1 \
    --outFileNamePrefix ${n}_unsorted. \
    --outSAMattributes All --outStd BAM_Unsorted --outSAMtype BAM
Unsorted \
    --outFilterType BySJout --outReadsUnmapped Fastx \
    --outFilterScoreMin 10 --outSAMattrRGline ID:foo --alignEndsType
EndToEnd >${n}_unsorted.bam
#比对病毒基因组序列
for i in *.out.mate1;do
    n=${i/.out.mate1/}
    STAR --runMode alignReads --runThreadN 9 --genomeDir ${index} \
        --genomeLoad LoadAndRemove --readFilesIn ${n}.out.mate1 \
        --outSAMunmapped within --outFilterMultimapNmax 1 --
outFilterMultimapScoreRange 1 \
        --outFileNamePrefix ${n}_unsorted. \
        --outSAMattributes All --outStd BAM_Unsorted --outSAMtype BAM
Unsorted \
        --outFilterType BySJout --outReadsUnmapped Fastx \
        --outFilterScoreMin 10 --outSAMattrRGline ID:foo --alignEndsType
Local >${n}_unsorted.bam
```

8. 对第7步每一步都要做一下处理，首先是对bam进行排序

```
gatk --java-options "-XX:+UseParallelOldGC -XX:ParallelGCThreads=4 -
XX:GCTimeLimit=50 -XX:GCHeapFreeLimit=10" SortSam --INPUT=${n}.bam --
OUTPUT=${n}_sorted.bam --VALIDATION_STRINGENCY=SILENT -SO=coordinate --
CREATE_INDEX=true
```

9. 其次是利用umitools对bam进行去除PCR重复

```
umi_tools dedup -I ${n}_sorted.bam --output-stats=deduplicatd -S
${n}_rmRep_sorted.bam
```

10. 并再次进行bam文件排序

```
gatk --java-options "-XX:+UseParallelOldGC -XX:ParallelGCThreads=4 -  
XX:GCTimeLimit=50 -XX:GCHeapFreeLimit=10" SortSam --  
INPUT=${n}_rmRep_sorted.bam --OUTPUT=${n}_rmRep_gatksorted.bam --  
VALIDATION_STRINGENCY=SILENT -SO=coordinate --CREATE_INDEX=true
```

11.将排序好的bam文件利用clipper进行call peak

```
clipper -b ${n}_rmRep_gatksorted.bam -s hg19 -o ${n}.peaks.bed --processors 10
```

12.利用软件对peak文件进行注释，可以用homer或者R语言，以下列出homer的使用

```
/home/gongweikang/.conda/envs/chipseq/share/homer/bin/annotatePeaks.pl  
${n}_c3.0_cond1.bed hg38 > ${n}_annotation.xls
```